MODEL comparision matrix

Classfication	Accuracy	Precision	Recall	F1Score
Logistic Regression	0.8468	0.7179	0.5995	0.6534
Decision Tree Classifier	0.8098	0.6027	0.6165	0.6095
Random Forest Classifier	0.8456	0.7097	0.6071	0.6544
Gradient Boosting Classifier (e.g., XGBoost or LightGBM)	0.8610	0.7686	0.6046	0.6768
K-Nearest Neighbors Classifier	0.8241	0.6574	0.5629	0.6065
Support Vector Classifier (SVC)	0.6579	0.3897	0.7432	0.5113
Voting vs Average (best 3 out of 6)	NA	NA	NA	NA
Ensemble	NA	NA	NA	NA

Which model gave you the best result and which was the worst?

Based on the results in the comparison table, Gradient Boosting performed the best overall, achieving the hig score and ROC-AUC on both the validation and test sets. This indicates strong predictive performance and goo generalization.

The Voting Ensemble also performed well, closely matching Gradient Boosting. By combining multiple models provided stable and consistent predictions, although it did not outperform the best individual model.

Random Forest was another strong performer, with high ROC-AUC and balanced F1-score, making it a reliable alternative.

Logistic Regression performed well on the validation set but showed poor generalization on the test set, sugg limitations in capturing complex patterns in the data.

KNN and SVC performed poorly, particularly in recall, indicating difficulty identifying positive cases. Naive Bay underperformed due to its assumption of feature independence, which is not realistic for this dataset.

Based on overall performance, Gradient Boosting is the recommended model for this task. It provides the bes of accuracy, F1-score, and ROC-AUC. The Voting Ensemble is a strong alternative, but did not significantly imp performance over Gradient Boosting.

Regression	MAE	MSE	Rsquared
Linear Regression			
Decision Tree Regressor			
Random Forest Regressor			
Gradient Boosting Regressor (e.g., XGBoost or LightGBM)			
K-Nearest Neighbors Regressor			
Voting Vs Avearge (best 3 out of 6)			
Ensemble			

Which model gave you the best result and which was the worst?

N/A – This project is a classification problem. Regression metrics such as MAE, MSE, and R^2 are not applicable

ROC/AUC
0.9002
0.7438
0.8986
0.9153
0.8392
0.759
NA
NA

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